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PERFORMANCE SPECIFICATION

GREASE, MULTIPURPOSE, WATER RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of grease for multipurpose usage in applications requiring water resistance. It is to be used at temperatures ranging from -18 to 110 °C (0 to 230 °F) in all shipboard applications requiring dispensing through long lengths of tubing. The temperature range for use in ball and roller bearing applications is -29 to 121 °C (-20 to 250 °F).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791 - Testing Method of Lubricants, Liquid Fuels, and Related Products

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual

(Copies of the chapter titled “Material Control Program” are available by email request to CommandStandards@navy.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering [hard copy] via the Naval Logistics Library [NLL] at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D217 - Standard Test Methods for Cone Penetration of Lubricating Grease

ASTM D942 - Standard Test Method for Oxidation Stability of Lubricating Greases by the Oxygen Pressure Vessel Method

ASTM D1092 - Standard Test Method for Measuring Apparent Viscosity of Lubricating Greases

ASTM D1264 - Standard Test Method for Determining the Water Washout Characteristics of Lubricating Greases

ASTM D1478 - Standard Test Method for Low-Temperature Torque of Ball Bearing Grease

ASTM D1743 - Standard Test Method for Determining Corrosion Preventive Properties of Lubricating Greases

ASTM D2265 - Standard Test Method for Dropping Point of Lubricating Grease Over Wide Temperature Range

ASTM D2596 - Standard Test Method for Measurement of Extreme-Pressure Properties of Lubricating Grease (Four-Ball Method)

ASTM D4048 - Standard Test Method for Detection of Copper Corrosion from Lubricating Grease

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D6184 - Standard Test Method for Oil Separation from Lubricating Grease (Conical Sieve Method)

ASTM E29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. For purposes of determining conformance with each requirement, an observed value or calculated value shall be rounded off to the nearest unit in the last right-hand place of figures used in expressing the limiting value in accordance with the rounding method in ASTM E29.

3.2 Qualification. The grease furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.3 Material. The grease shall consist of a liquid lubricant and gelling agent that when formulated, results in a compound meeting this specification. The grease shall not contain additives of the solid type.

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3.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Toxicity and prohibited materials.

3.5.1 Toxicity. When evaluated in accordance with 4.5.1, the grease shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.1 and 6.6).

3.5.2 Prohibited materials. The grease shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

3.6 Off-gassing. The grease shall be evaluated for off-gassing in accordance with the requirements of 4.5.3. Based on the circumstances of use and the chemical nature of the grease, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the grease shall be assigned to either the "Permitted" or "Limited" category (see 4.5.3 and 6.5).

3.7 Ozone depleting substances (ODS). The use of ODSs in the composition of the grease under this specification directly or reference in any federal test method is hereto prohibited. Environmentally safe and non-ODS alternative solvents shall be substituted in test methods for any previously specified ODS such as 1,1,1-trichloroethane, tri-chlorofluoroethane, perchloroethane, or other chlorofluorocarbons. A dehydrated mixture of primary alcohol, such as ethanol or isopropyl alcohol, and non-aromatic hydrocarbon solvent resulting in a minimum flash point of 60.0 °C (140.0 °F) plus effective solvent action shall be used as an alternative for any ODS.

3.8 Odor. The grease shall not have an objectionable odor or odor of rancidity, perfume, or free alcohol (see 4.5.5).

3.9 Workmanship. The grease, when examined visually, shall be a smooth and homogenous mixture, free from lumps and extraneous materials (see 4.5.6).

3.10 Compatibility. The grease shall be compatible with all previously qualified greases when tested in accordance with 4.5.7. Immediately after mixing, and after undisturbed storage for 7 days, the mixture shall not form resinous gums, sludge, or insoluble solid materials and shall comply with the worked penetration and penetration after prolonged working requirements (see table I). In the event of failure, the previously qualified grease (unmixed with other greases) shall be tested to verify that it meets the worked penetration and worked stability requirements under the conditions specified in 4.5.7. If the previously qualified sample fails to meet the worked penetration or worked stability requirement, a fresh sample shall be obtained and the compatibility test shall be repeated.

3.11 Chemical and physical properties. Chemical and physical properties shall be as specified in table I.

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TABLE I. Chemical and physical requirements and test methods.

Property	Requirement	Test Method (ASTM or Other) or Verification Paragraph
Dropping point, °C (°F), min	149 (300)	ASTM D2265
Penetration, worked (60 double strokes)	265-320	ASTM D217
Penetration of lubricating greases after prolonged working (after 100,000 double strokes), max 1/	355	ASTM D217
Apparent viscosity, max 2/	250 poises at 200 inverse seconds	ASTM D1092
Oil separation after 30 hours at 100±3 °C (212±5 °F), % max	Report	ASTM D6184
Dirt content, particles per mL grease, max:		FED-STD-791, method 3005
25-75 µm 3/	7500	
75-125 µm	1600	
Over 125 µm	0	
Corrosiveness (copper strip), rating, max	1b	ASTM D4048
Oxidation stability, pressure drop, kilopascals (pounds per square inch), max		ASTM D942
After 100 hours	69 (10)	
After 500 hours	172 (25)	
Water washout at 38 °C (100 °F), % max	5.0	ASTM D1264
Load wear index, kilogram-force, min	30	ASTM D2596
Low temperature torque at -29±1 °C (-20±2 °F), newton-meter (grams-centimeter), max		ASTM D1478
Start	0.441 (4500)	
Running, after 1 hour	0.147 (1500)	
Rust preventive properties, rating	Pass	ASTM D1743
Storage stability, penetration:		4.5.8.1
Unworked, min	200	
Worked (compared to original), max	±30	
Separation of crystalline material (6 months)	None	
Performance test, hours, min 4/	2000	4.5.8.2

NOTES:

1/ Test may be referred to as “worked stability” in some publications.

2/ Test temperature: 0±0.5 °C (32±1.0 °F). The grease shall be cooled to test temperature over the course of 2 hours and shall be maintained at temperature for 2 hours prior to making the viscosity determination.

3/ Micrometer (µm).

4/ Information about approved test facilities can be obtained by contacting NAVSEA (see 6.3).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the conditions specified in table I on unworked grease. The physical values specified in table I apply to the average of determinations made on the sample.

4.3 Qualification inspection. Qualification tests shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall consist of all the tests specified in the qualification test column of table II to satisfy the requirements in section 3 and table I (see 6.3).

4.3.1 Sampling. Qualification test samples shall consist of a minimum of 15.0 lb (6.81 kg) of grease.

4.3.2 Change approval. A change in product composition, formulation, or ingredients used in the manufacture of grease that has been qualified shall require written approval of NAVSEA. Incorporation of any changes may require requalification of the item in question.

4.3.3 Requalification. Unless otherwise approved by NAVSEA, requalification shall consist of all the tests specified in the qualification test column of table II to satisfy the requirements in section 3 and table I (see 6.3).

4.4 Conformance inspection. The conformance inspection shall consist of the all the tests specified in the conformance test column of table II. The inspection shall be conducted on samples taken in accordance with 4.4.1.

4.4.1 Sampling for tests. Two 2.95-kg (6.5-lb) containers of grease shall be selected, taken at random from each filled lot (see 6.7.2). For users who obtain grease in large containers, samples shall be selected in accordance with ASTM D4057. One sample shall be used for tests and the other shall be retained. If any test result does not comply with the stated requirement, the test in question shall be repeated on the retained sample. If the sample fails again, the lot shall be rejected.

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TABLE II. Qualification tests and conformance tests.

Tests	Requirement	Verification	Qualification Test	Conformance Test
Toxicity	3.5.1	4.5.1	X	
Prohibited materials	3.5.2	4.5.2	X	
Off-gassing	3.6	4.5.3	X	
ODSs	3.7	4.5.4	X	
Odor	3.8	4.5.5	X	X
Workmanship	3.9	4.5.6	X	X
Compatibility	3.10	4.5.7	X	
Dropping point	3.11	4.5.8	X	X
Penetration, worked	3.11	4.5.8	X	X
Penetration after prolonged working	3.11	4.5.8	X	
Apparent viscosity	3.11	4.5.8	X	X
Oil separation	3.11	4.5.8	X	X
Dirt content	3.11	4.5.8	X	X
Corrosiveness (copper strip)	3.11	4.5.8	X	X
Oxidation stability	3.11	4.5.8	X	
Water washout	3.11	4.5.8	X	
Load wear index	3.11	4.5.8	X	
Low temperature torque	3.11	4.5.8	X	X
Rust preventive properties	3.11	4.5.8	X	
Storage stability	3.11	4.5.8.1	X	
Performance test	3.11	4.5.8.2	X	

4.5 Test methods.

4.5.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.5.1 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the grease using data provided by the manufacturer/distributor to the NMCPHC (see 3.5.1 and 6.6).

4.5.2 Prohibited materials. Prohibited material shall be verified for conformance to 3.5.2 as required by the qualifying activity.

4.5.3 Off-gassing. The grease shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled “Material Control Program” (see 3.6 and 6.5). If the Navy determines that off-gas testing is required, testing shall be conducted at a NAVSEA approved test facility (see 3.6). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.6).

4.5.4 ODSs. Verification that an ODS is not used in the formulation of the grease or testing thereof shall be accomplished as specified by the qualifying activity (see 3.7).

4.5.5 Odor. The odor of the grease shall be observed immediately upon opening the test container (see 3.8).

4.5.6 Visual examination. Visual examination shall be conducted to determine conformance with the requirements of 3.9.

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4.5.7 Compatibility. Standard grease cups shall be filled with equal volumes of the grease being tested and previously qualified greases. The cup shall be filled by alternately adding 60 grams of each of the two greases being tested, until the cup is full (approximately 454 grams). The two greases shall be mixed by working the sample for 60 double strokes in accordance with the worked penetration procedure of ASTM D217, and then let the mixture stand undisturbed for 7 days at 18.0 to 29.5 °C (65 to 85 °F) and 50±10 percent relative humidity. The worked penetration and the penetration after prolonged working shall be in accordance with table I, using the same sample of grease mixture for both tests. Small amounts of fresh grease may be added to the worker cup just prior to the worked stability test as needed to maintain a full cup for testing.

4.5.8 Chemical and physical properties. The chemical and physical property tests shall be performed in accordance with the applicable method of FED-STD-791, ASTM standards, or applicable test paragraphs as shown in table I.

4.5.8.1 Storage stability. Two standard grease worker cups (ASTM D217) shall be filled with the grease sample, one struck flush with the top of the cup for undisturbed penetration, and the second filled with an additional quantity of grease to ensure sufficient sample to conduct the worked penetration. The sample for unworked penetration shall be covered with a watch glass. The sample for worked penetration shall be covered with aluminum foil. The cups shall be stored covered in an oven at a temperature of 38±3 °C (100±5 °F) for 6 months subsequent to the original determination of the penetration. Upon termination of storage the samples are allowed to cool to 25 °C (77 °F) prior to determining the undisturbed and worked penetration in accordance with ASTM D217. The undisturbed penetration sample shall be stored at 38±3 °C (100±5 °F) for an additional 6 months. The sample shall be covered in such a way as to provide an air space above the grease and prevent the surface of the grease from being disturbed. The sample shall then be examined for the presence of crystalline material.

4.5.8.2 Performance test.

4.5.8.2.1 Apparatus. The apparatus shall be as shown on figure 1 and shall consist of the following:

a. Test spindle supported in a cast iron housing by two auxiliary bearings.

b. Driving motor belted to the test spindle capable of rotating the spindle at a speed of 3600±50 revolutions per minute.

c. Electrically heated oven thermostatically controlled to maintain the temperature of the test bearing within ±1.7 °C (3 °F) of the chosen temperature.

d. Test bearing shall be single-row, radial, deep-groove, unshielded with pressed steel retainer, size 310, grade ABEC-1 with normal internal fit-up, made of SAE 52100 steel, enclosed in a housing, and maintained in a position on the test spindle by a light press fit. A radial load of 667.2 newtons (150 pounds) is applied to the bearing by a spring mounted outside the oven. Details of the bearing housing shall be as shown on figures 2 and 3.

e. Covered ultrasonic cleaner parts washer with heater for cleaning test bearing.

(1) Cleaning material. Electron solvent (NSN 6850-01-375-5553 [6-gallon pail], NSN 6850-01-375-5554 [1-gallon can]) or equivalent aliphatic hydrocarbon terpene mixture.

(2) Drying material. Hypersolve (NSN 6850-01-450-6162 [5-gallon pail]) or equivalent drying agent.

Note: All steps involving solvent and drying agent shall be accomplished inside a chemical fume hood with positive ventilation, while wearing solvent resistant gloves (e.g., Viton or equal) and safety glasses with splash protection.

4.5.8.2.2 Preparation. For pregreased test bearings, grease shall be removed from bearings using a narrow blade nonmetallic spatula. All test bearings shall be cleaned as follows:

a. Place the bearing into a parts washer and hand wash the bearing by slowly spinning it while partially submerged in the electron solvent. The solvent shall be used at room temperature since heating the solvent may cause the terpene to become unstable.

b. Manually remove stubborn deposits or clumps of grease with a clean brush having short, soft bristles.

c. Replace the solvent to fully cover the bearings.

d. Periodically inspect the bearing. When the visible face of the bearing is clean, rotate the bearing slightly.

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e. Quickly dip the bearing in a container filled with clean solvent; quickly dip the bearing in a stainless steel container filled with drying material; place the bearing on a clean dry surface until all of the residual materials have evaporated.

f. The bearing is weighed.

g. Grease is injected into the bearing using a grease gun with a needle tip having a 3/32-inch inside diameter until 30±0.1 grams of grease having a specific gravity of 0.9 are uniformly distributed throughout the bearing. Grease packs for grease with specific gravity other than 0.9 are as follows:

$$Grease\ pack\ (gram) = \frac{30\ x\ (specific\ gravity\ of\ grease)}{0.9}$$

Grease shall not extend beyond the face of the races. The bearing shall then be installed in the bearing housing.

4.5.8.2.3 Procedure. On initial start-up, the testers shall be run for 24 hours without heaters. Then the heaters shall be turned on and sufficient heat applied to bring the test bearing to the specified temperature of 121±3 °C (250±5 °F) within 2 hours. After 144 hours of operation with heat, the spindle shall be stopped and the heat turned off. After a 24-hour shutdown, the heated operating cycle shall be repeated. This operating cycle shall be continued until the specified number of hours has been attained or until lubrication failure occurs. During each shutdown, the bearing housing shall be inspected for grease leakage, shown by grease on the face of the bearing housing or on the rod supporting the radial load. Failure of lubrication shall be considered to have occurred when any of the following conditions prevail:

a. Tripping of motor thermal cut-out switch indicating that the frictional torque has increased sufficiently to overload the motor.

b. Locking of test bearing and belt slippage at start of test run.

c. Excessive leakage shown by flow of grease or oil down the face of the test bearing housing or radial load rod.

The test shall be conducted on four bearings. The lubricant shall be reported as passing the test if three of the four bearings each operate satisfactorily for the number of hours specified.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The grease covered by this specification is intended for multipurpose use in applications requiring water resistance in all shipboard applications requiring dispensing through lengths of tubing at temperatures from -18 to 110 °C (0 to 230 °F); and in ball and roller bearings at temperatures from -29 to 121 °C (-20 to 250 °F). This grease has unique physical and chemical requirements that are compatible for use across the Navy. Current stock of the grease may be used until depleted; future requisitions for the grease should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required.

c. When inspections should not be performed on unworked grease in accordance with the conditions specified in table I on (see 4.2).

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d. Packaging requirements (see 5.1).

e. Activities requiring copies of completed safety data sheets (SDS) (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24139 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Copies of SD-6, "Provisions Governing Qualification," are available online at https://quicksearch.dla.mil/ or https://assist.dla.mil.

6.3.2 Submission of qualification samples and additional information. When authorizing the forwarding of qualification samples, the qualifying activity will require the manufacturer to submit for review and approval, along with the samples, the manufacturer's test report (which includes the sample's SDS (see 6.4), the location and identity of the plant which produced the sample, and the item composition report. The samples should be plainly and durably marked with the following information:

a. "Sample for Qualification Test"

b. "GREASE, MULTIPURPOSE, WATER RESISTANT"

c. "Specification MIL-PRF-24139"

d. Manufacturer's name

e. Manufacturer's phone number

f. Product code number

g. Batch (see 6.7.1) or lot number

h. Date of manufacture

i. "Submitted by (name and date), for qualification inspection in accordance with MIL-PRF-24139, under authorization of (reference authorization letter)"

6.4 SDSs. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.5 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

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6.6 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/sites/nmcphc/industrial-hygiene/pages/health-hazard-assessment.aspx.

6.7 Definitions.

6.7.1 Batch. A batch is the quantity of material that has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.7.2 Lot. A lot consists of the grease produced by one manufacturer at one plant, from the same materials and under essentially the same conditions, provided the operation is continuous and does not exceed a 24-hour period.

6.8 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, DoD Shelf-Life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.9 Subject term (key word) listing.

High dropping point

Lubricant

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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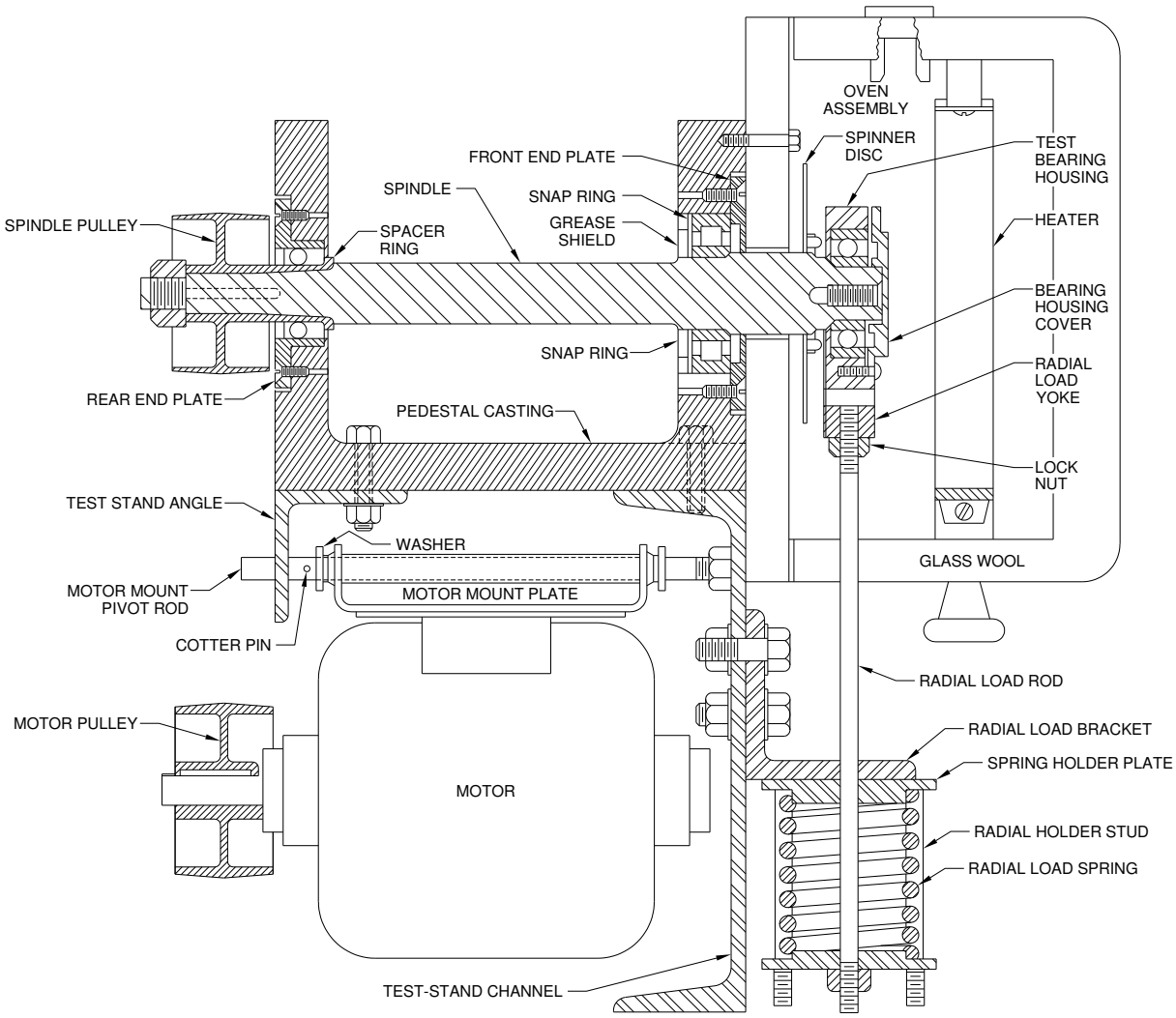


FIGURE 1. Performance tester for electric motor greases.

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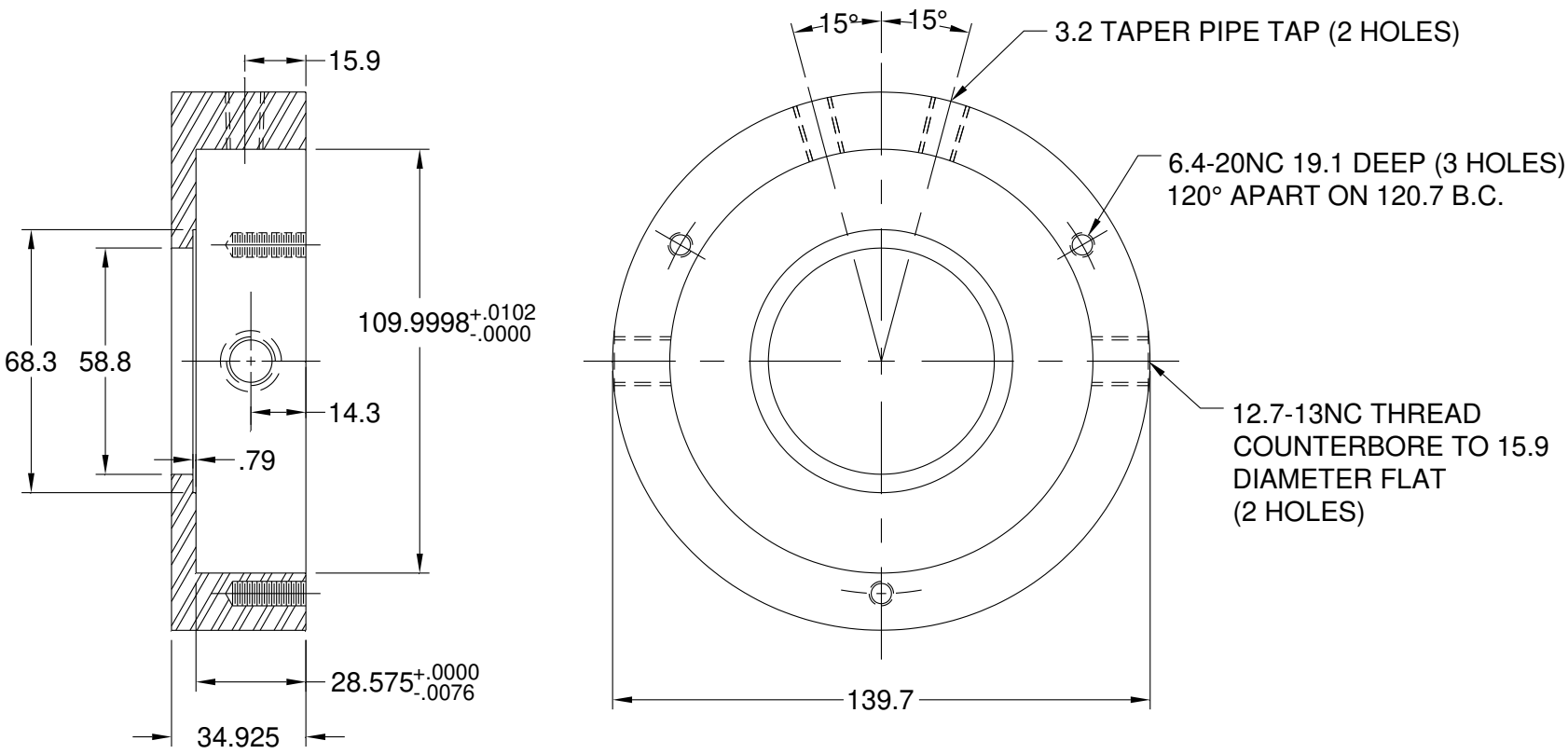


FIGURE 2. Test bearing housing.

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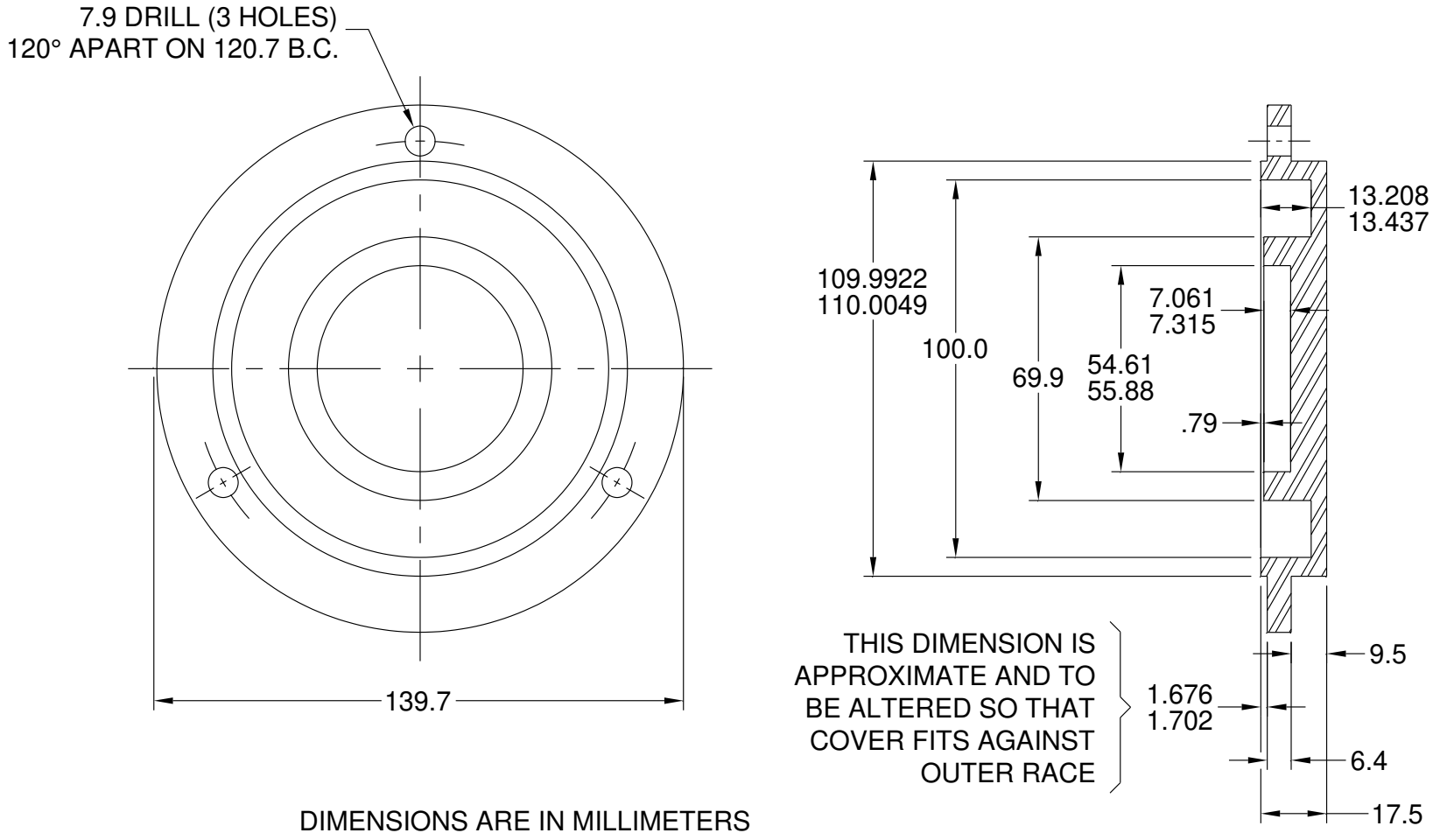


FIGURE 3. 310 bearing house cover.

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CONCLUDING MATERIAL

Custodians:
Navy – SH
Air Force – 68
DLA – GS

Review activities:
Navy – AS
DLA – PS

Preparing activity:
Navy – SH
(Project 9150-2020-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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